Particle A with mass 3M sits stationary in the lab frame. Suddenly it decays into two daughter particles: Particle B with mass 2M and traveling in the -x direction with velocity v = -0.5c, and particle C with mass M traveling in the +x direction with unknown speed. The purpose of this worksheet is to solve for the speed.

1. Draw this situation both before and after the decay.

2. Show that the four-momentum of particle A before the decay is

$$\mathbf{P_A} = \begin{bmatrix} E \\ p_x c \end{bmatrix} = \begin{bmatrix} 3mc^2 \\ 0 \end{bmatrix}$$

where E is A's total energy, $E = \gamma mc^2$ and $p_x = \gamma m v_x$ is its total relativistic x-momentum.

3. Show that $|\mathbf{P_A}|^2 = \mathbf{P_A} \cdot \mathbf{P_A} = -9m^2c^4$. What the invariant of $\mathbf{P_A}$?

After the collision there is the momentum of particle A and particle B. But the total momentum is their (four) vector sum. Simple algebra shows:

$$(\mathbf{P_B} + \mathbf{P_C}) \cdot (\mathbf{P_B} + \mathbf{P_C}) = |\mathbf{P_B}|^2 + 2\mathbf{P_B} \cdot \mathbf{P_C} + |\mathbf{P_C}|^2$$

Since the total momentum before and after the collision should be the same

$$\begin{aligned} |\mathbf{P_A}|^2 &= |\mathbf{P_B}|^2 + 2\mathbf{P_B} \cdot \mathbf{P_C} + |\mathbf{P_C}|^2 \\ -9m^2c^4 &= |\mathbf{P_B}|^2 + 2\mathbf{P_B} \cdot \mathbf{P_C} + |\mathbf{P_C}|^2 \end{aligned}$$

4. Show that

$$\mathbf{P_B} = \begin{bmatrix} E \\ p_x c \end{bmatrix} = \begin{bmatrix} \frac{4}{\sqrt{3}} mc^2 \\ -\frac{2}{\sqrt{3}} mc^2 \end{bmatrix}$$

5. Show that $|\mathbf{P_B}|^2 = -4m^2c^4$

6. Show that $|\mathbf{P_C}|^2 = \gamma^2 m^2 c^4(\beta^2 - 1) = -m^2c^4$

7. Show that $2\mathbf{P_B} \cdot \mathbf{P_C} = -\frac{4m^2c^4\gamma}{\sqrt{3}}(2 + \beta)$

8. Using the values you found on this page plug into the equation on the very bottom of page 1 to show that $\beta = \sqrt{3/7} \approx 0.654$. .